VEL SOFTWARE SOLUTIONS · VEL-CSE-001
Companion template to the Custom Software & Enterprise Solutions Department Handbook · v1.0 · 4 November 2026

Appendix F — Database Design Checklist

Modelling

- [ ] Entities and relationships trace to FR IDs.
- [ ] 3NF unless a documented, measured reason to denormalise.
- [ ] Naming convention applied (snake_case, singular tables, `_id` keys).
- [ ] Master vs transaction vs reference tables separated.

Integrity

- [ ] Every table has a primary key.
- [ ] Foreign keys exist for every relationship (with defined ON DELETE behaviour).
- [ ] UNIQUE constraints for natural keys (e.g. invoice number per period).
- [ ] CHECK constraints for ranges/statuses; NOT NULL by default.
- [ ] Money in `numeric/decimal(14,2)`, never float; timestamps in UTC (`timestamptz` / `datetime2`).

Concurrency & transactions

- [ ] Transaction boundary per use case documented.
- [ ] Row-version/optimistic concurrency on editable records.
- [ ] Gap-free number series use a locked counter row, not `MAX()+1`.

Performance

- [ ] Indexes on FKs and common filters; each justified by a query plan.
- [ ] No `SELECT *` in application code; pagination on all list queries.
- [ ] Reporting queries tested with production-scale data.

Security

- [ ] Least-privilege DB accounts (app ≠ migration ≠ admin).
- [ ] Sensitive columns identified; encryption at rest/in transit; masking in non-production.
- [ ] Audit table append-only (no UPDATE/DELETE grants).
- [ ] If multi-tenant is *explicitly required*: tenant_id + isolation tests.

Migrations

- [ ] Forward-only versioned scripts; applied scripts never edited.
- [ ] Rehearsed on production-sized copy; lock duration measured.
- [ ] Rollback = restore or compensating migration documented.

Backup & recovery

- [ ] Backup schedule and retention documented; off-site encrypted copy.
- [ ] RPO/RTO agreed with client.
- [ ] **Restore tested** and recorded (date, duration, result).

Documentation

- [ ] ERD current; data dictionary lists each column's meaning, type and source.

Approval

Role	Name	Signature	Date
Database Engineer			
Technical Lead			